



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Student Name: Y Jayanth Reddy

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Course Code: CSA0211

Slot: B

Course Name: C-PROGRAMMING

Course Faculty: Dr. K. S. SARAIVANAN

Project Title: Smart University Laboratory Equipment Monitoring and Allocation System using C

Module Photographs:



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1. Add Equipment
2. Update Equipment
3. Display All Equipment
4. Search Equipment (ID / Category / Lab / Condition / Availability)
5. Sort Equipment (ID / Category / Usage / Availability)
6. Merge Records from Another Laboratory File
7. Analyse Equipment Status (Available/Low/Maintenance/Critical)
8. Allocate Equipment to a Requesting Lab/User
9. Generate Consolidated Report
10. Save Records to File
0. Exit

Enter your choice: 0
Data auto-saved. Exiting. Goodbye!

...Program finished with exit code 0
Press ENTER to exit console.
```

Project Description: (here you write what you did in this project (contribution) including Model Description)

The project titled “**Smart University Laboratory Equipment Monitoring and Allocation System using C**” was developed to manage and monitor laboratory equipment efficiently. The system allows users to add, update, display, search, sort, and allocate laboratory equipment. It also monitors equipment availability, condition, usage count, and maintenance status. The project uses C programming concepts such as structures, arrays, functions, pointers, searching, sorting, recursion, and file handling. The model helps identify low-availability and critical equipment, supports equipment allocation, and generates consolidated reports. My contribution included developing the core program logic, implementing equipment management and allocation features, searching and sorting operations, status analysis, and file-based data storage.

Student Signature

Guide Signature